

Airframe Technology Master

FATIGUE & DT

Lecture 14

Damage Tolerance. Residual Strength

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Damage tolerance

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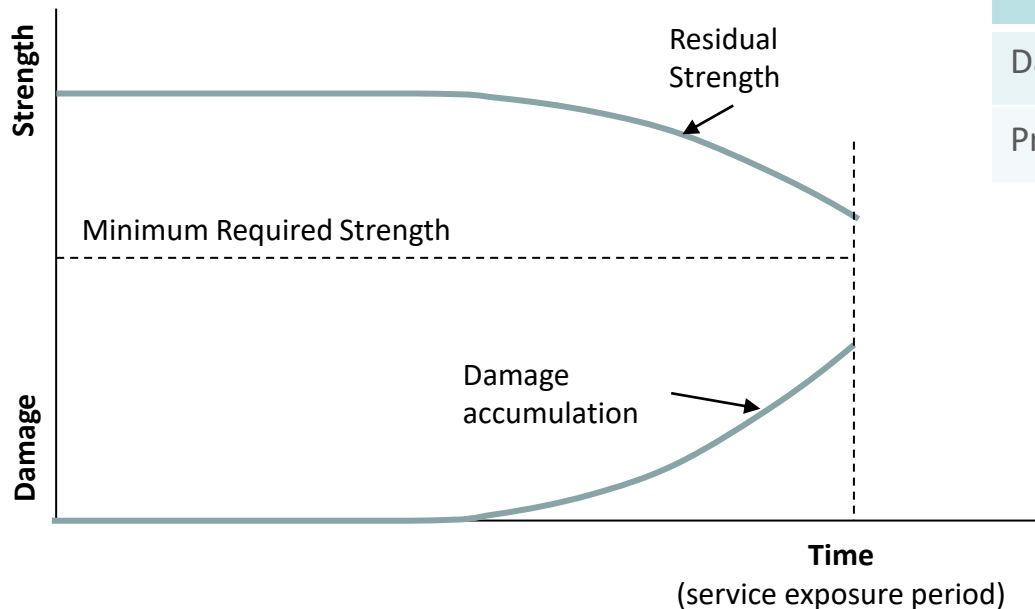
Damage Tolerance. Residual Strength

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Scope

What is residual strength?

The residual strength capability is defined as the amount of static strength available at any time during the service exposure period considering that damage is initially present and grows as a function of service exposure time (USAF Handbook ref. [1]).



Strength degradation due to:

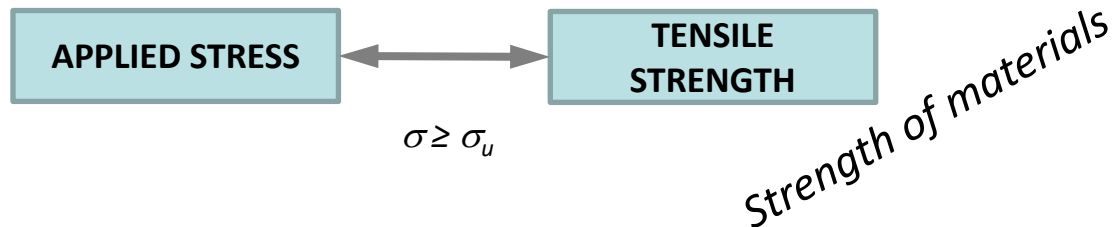
Parameter	Metallic	Composite
Damage	Growth	No growth
Properties	No degradation	Degradation

This lecture is focused on metallic materials.

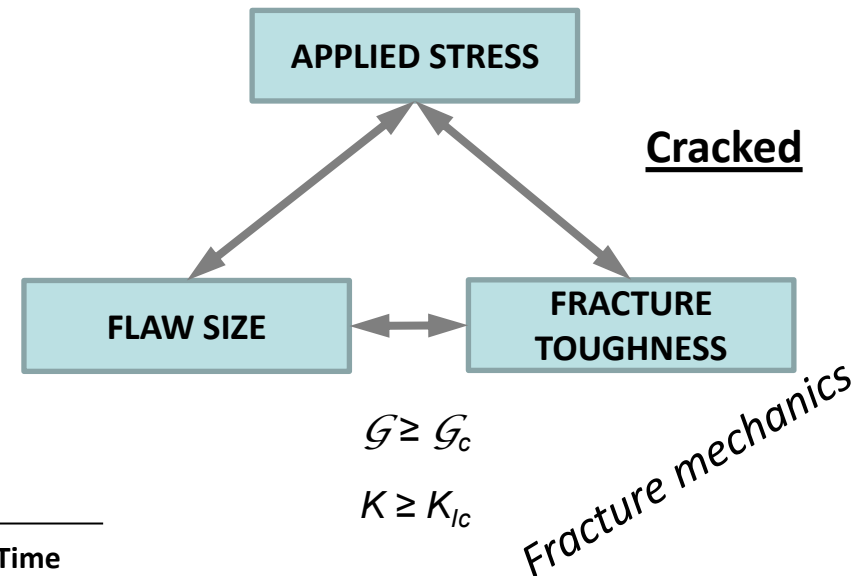
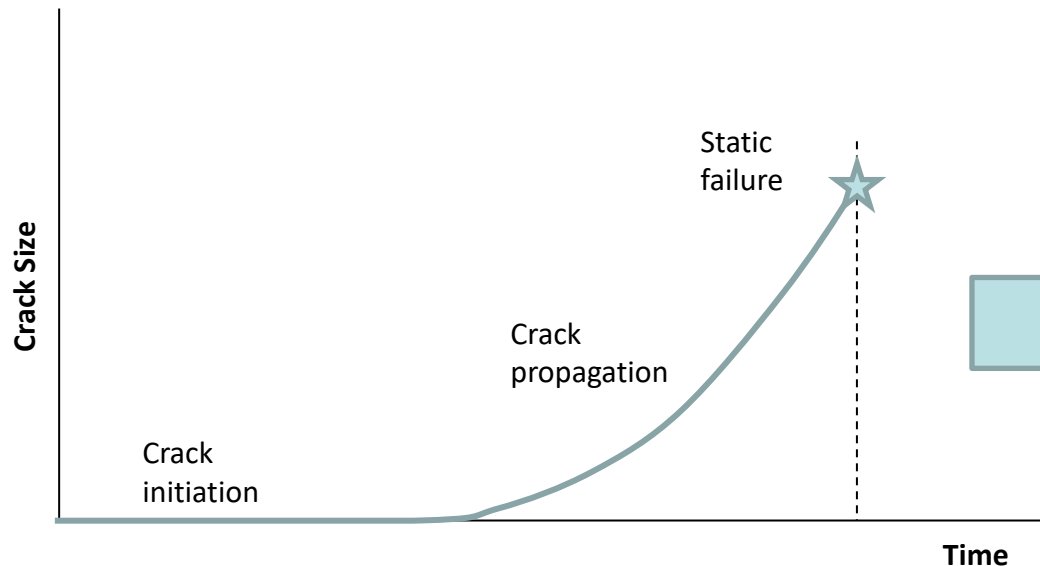
Phases of Failure in Metallic Material

Un-cracked vs cracked structure:

Un-cracked



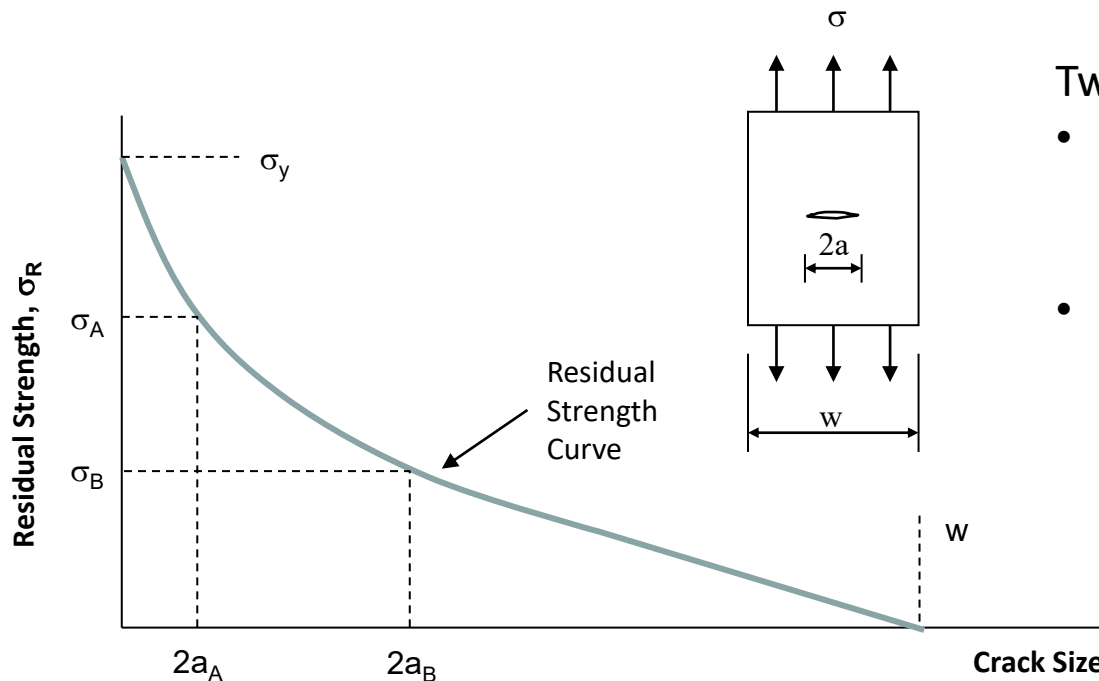
This lecture is focused on the analysis of rupture of cracked metallic component under the application of a static load.



Residual Strength Curve

The residual strength of a cracked component depends on several parameters:

- Parameters inherent to the configuration (geometry, materials, assembly, etc)
- Loading mode (tension, shear, etc)
- Crack size



Two points of view:

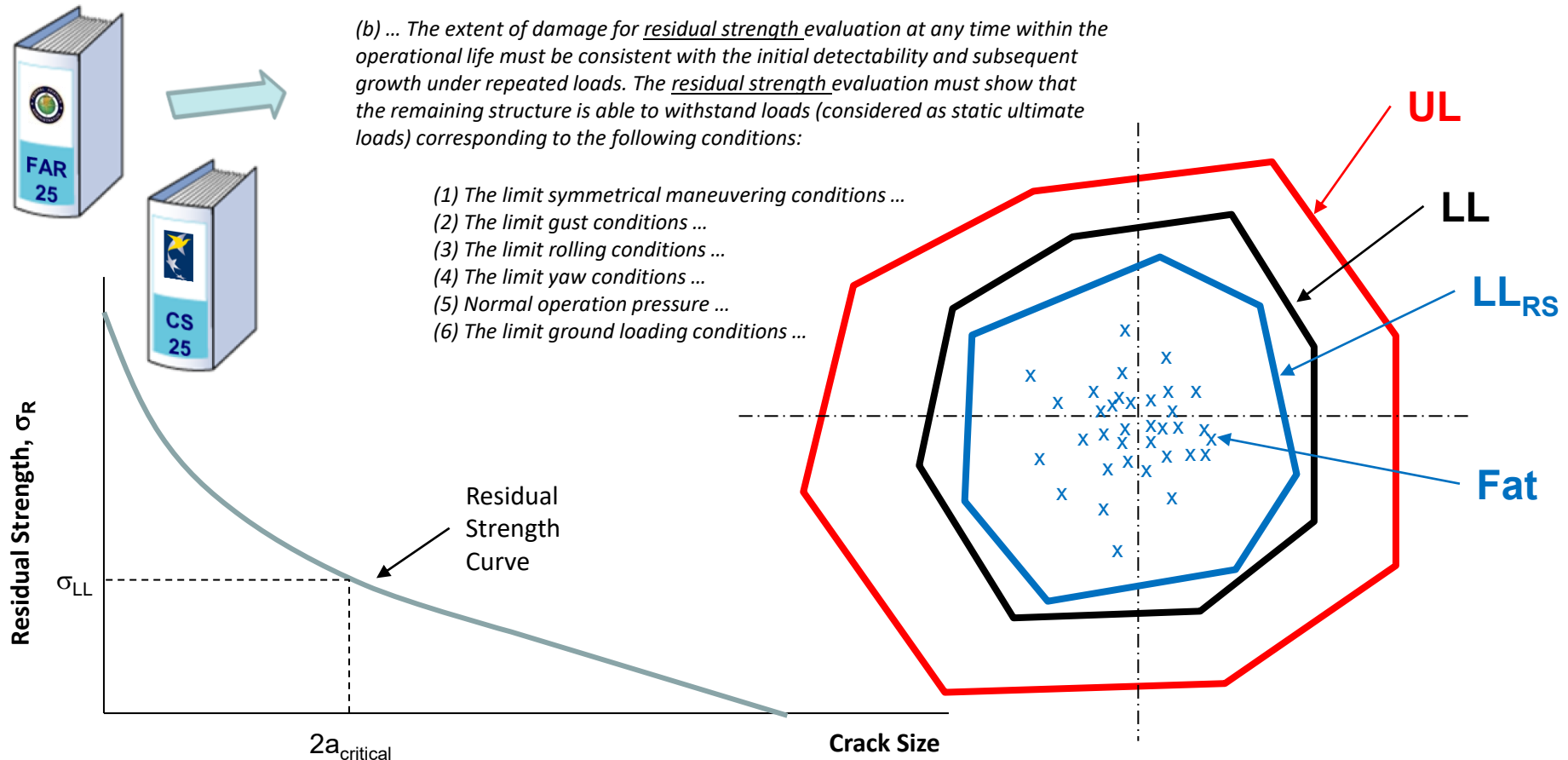
- It allows evaluating the maximum load or stress sustainable with a given crack size
- It allows evaluating the maximum crack size allowed for a certain load or stress level

Compliance with Airworthiness Regulation

Basic regulation for airplanes: FAR 25.571 (FAA), CS 25.571 (EASA).

(b) ... The extent of damage for residual strength evaluation at any time within the operational life must be consistent with the initial detectability and subsequent growth under repeated loads. The residual strength evaluation must show that the remaining structure is able to withstand loads (considered as static ultimate loads) corresponding to the following conditions:

- (1) The limit symmetrical maneuvering conditions ...
- (2) The limit gust conditions ...
- (3) The limit rolling conditions ...
- (4) The limit yaw conditions ...
- (5) Normal operation pressure ...
- (6) The limit ground loading conditions ...



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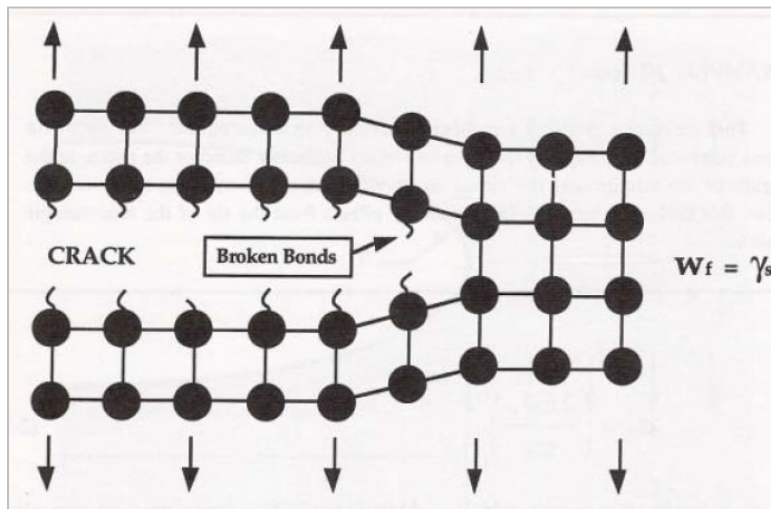
Brittle Materials

Energy criterion for fracture:

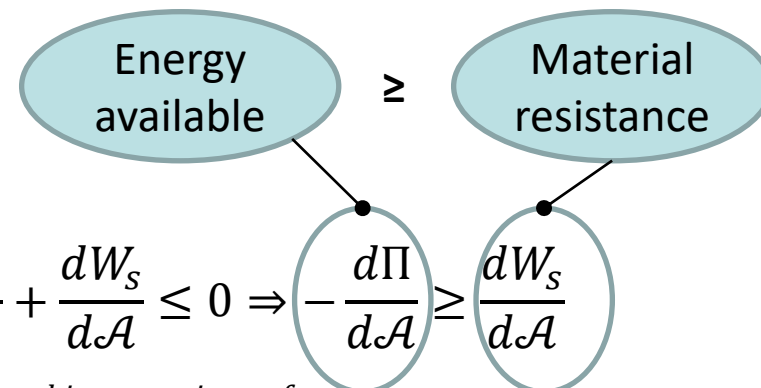
- Developed by A.A. Griffith in the 1920's (ref. [2])
- Motivation: match theoretical strength of glass ($\approx 100,000$ MPa to break atoms bonds) with real strength (≈ 100 MPa)
- Criterion: crack extension (i.e. fracture) occurs when the energy available for crack growth is sufficient to overcome the resistance of the material (atoms bonds)



A.A. Griffith (1893-1963)



From ref. [3]



$$\frac{dE}{d\mathcal{A}} = \frac{d\Pi}{d\mathcal{A}} + \frac{dW_s}{d\mathcal{A}} \leq 0 \Rightarrow -\frac{d\Pi}{d\mathcal{A}} \geq \frac{dW_s}{d\mathcal{A}}$$

$d\mathcal{A}$ = incremental increase in surface area

E = total energy

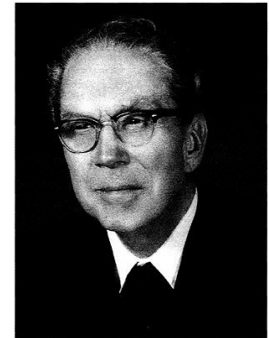
Π = potential energy: internal strain and external forces

W_s = work required to create new surfaces

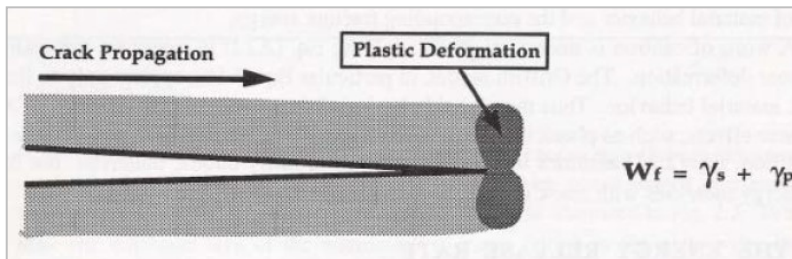
Ductile Materials

Energy criterion for fracture:

- Developed by G.R. Irwing in the 1950's (ref. [4])
- Motivation: extension of Griffith method, since it underestimates the fracture strength of metals (with small plastic deformation).
- Criterion: crack extension (i.e. fracture) occurs when the energy available for crack growth is sufficient to overcome the resistance of the material (atoms bonds and plastic work)



G.R.. Irwing (1907-1998)



From ref. [3]

Energy release rate: rate of change in potential energy with crack area (measure of the energy available for an increment of crack extension).

$$\mathcal{G} = -\frac{d\Pi}{d\mathcal{A}}$$

$$\mathcal{G}_c = \frac{d(W_s + W_p)}{d\mathcal{A}}$$

Energy release rate $\geq$ Critical $\mathcal{G}$

$\mathcal{G} \geq \mathcal{G}_c$

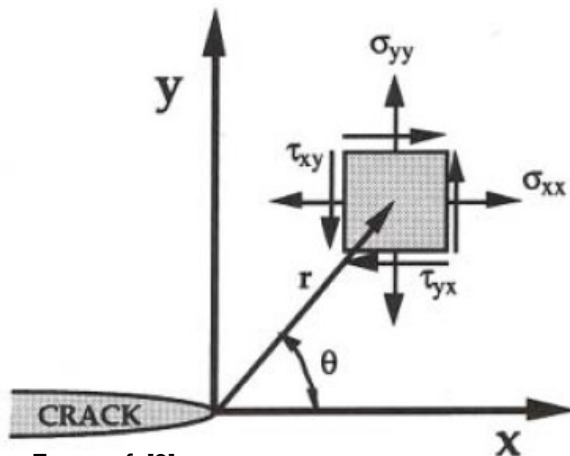
Ductile Materials

Stress intensity criterion for fracture:

- Developed by G.R. Irwing in the 1950's (ref. [6])
- Analysis: Irwing used the Westergaard approach for stress analysis ahead of a crack to show that the stresses and displacements near the crack tip could be described by a single constant related to the energy release rate, latter known as stress intensity factor K.
- Criterion: fracture occurs when the stress intensity factor K is greater than the fracture toughness of the material K_c .



H.M. Westergaard (1888-1950)



From ref. [3]

$$\sigma_{ij} = \frac{K}{\sqrt{2\pi r}} f_{ij}(\theta)$$

Stress intensity factor

≥

Fracture toughness

K

≥

K_c

Different configurations with same K implies same local stress-displacement field, so same material resistance (K_c).

Other approaches

The approaches described previously, based on stress intensity factor, form the bases of linear elastic fracture mechanics, where plastic zone size is small.

When plastic zone is not small, non-linear fracture mechanics is used, such as:

- J-Integral
- CTOD (Crack Tip Opening Displacement)
- Others

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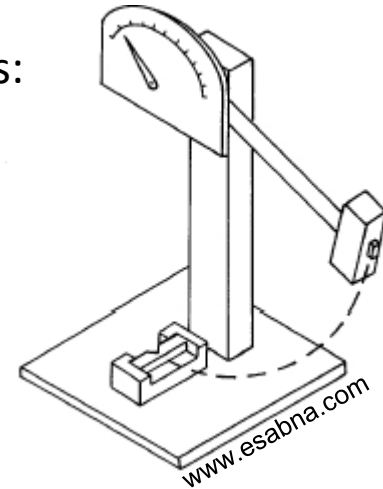
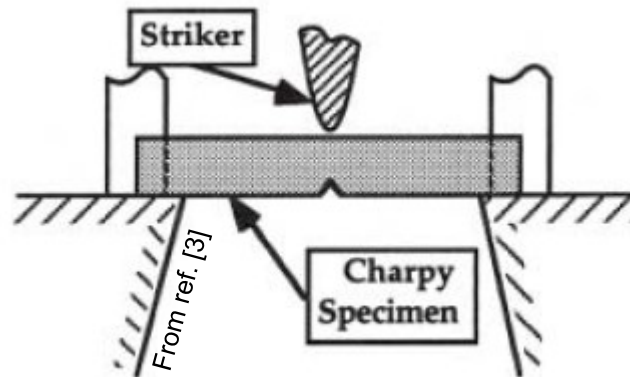
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 - Factors / parameters affecting fracture toughness
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Test of Fracture Toughness

Several tests have been design for the assessment of fracture toughness:

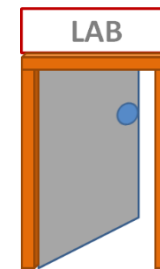
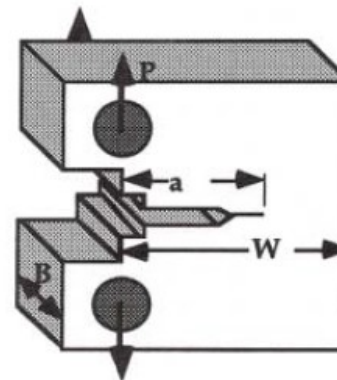
- Charpy test

It is a qualitative test, design for measuring fracture energy. It is used for qualification of materials, it is not useful for analysis.

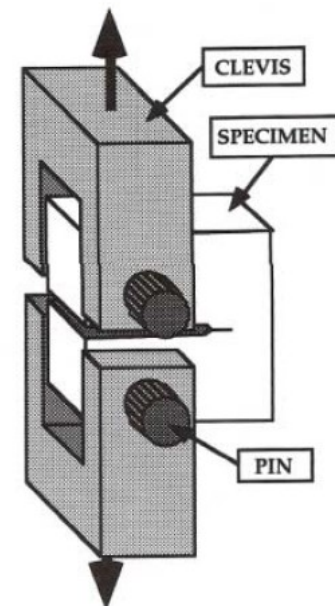


- K_{IC} test

It is adequate for stress analysis using stress intensity factor methodology.



From ref. [3]



- Some other tests will be presented in the following slides

Factors / Parameters Affecting Fracture Toughness

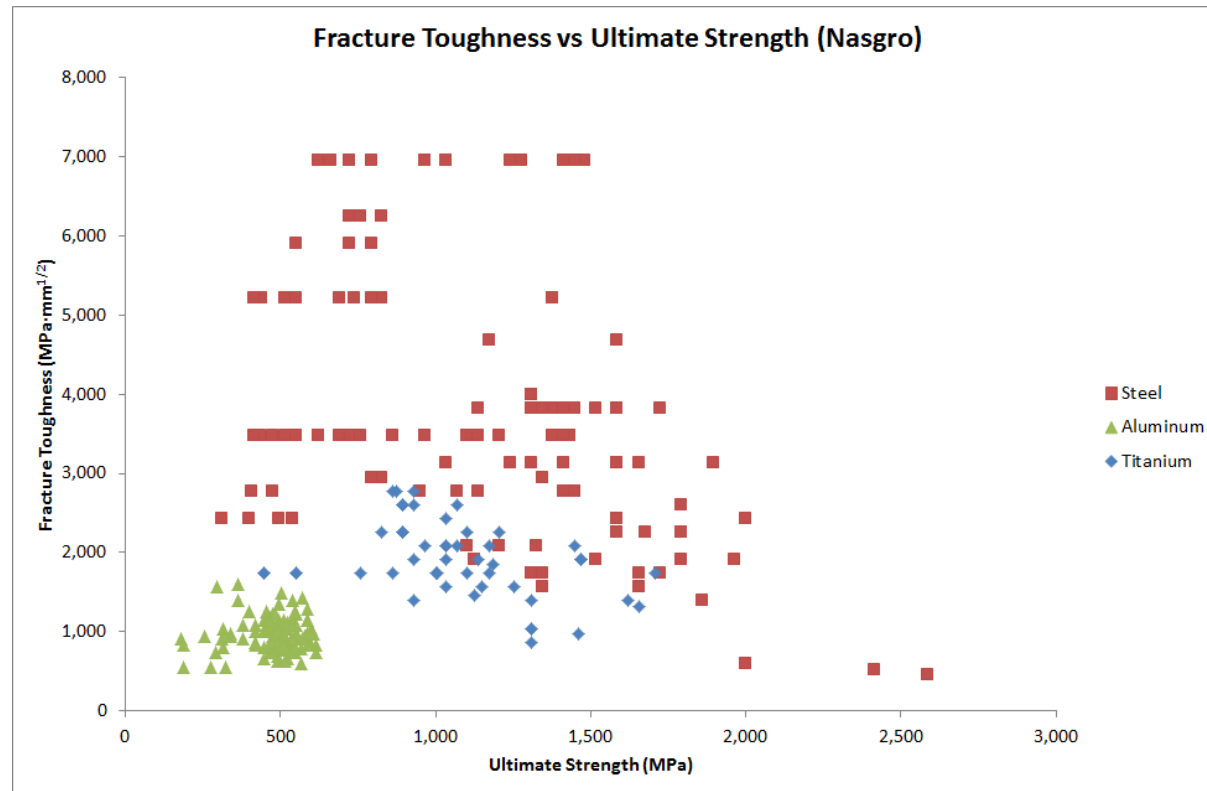
Fracture toughness K_c depends on:

- The material, heat treatment, product form
- Temperature
- Environmental conditions

As indicated previously, the stress intensity factor K is a parameter that defines the stress-displacement field on the crack tip area.

Then, for the same material, different geometrical configurations and stress levels but with the same K would fail at the same K value, called K_c .

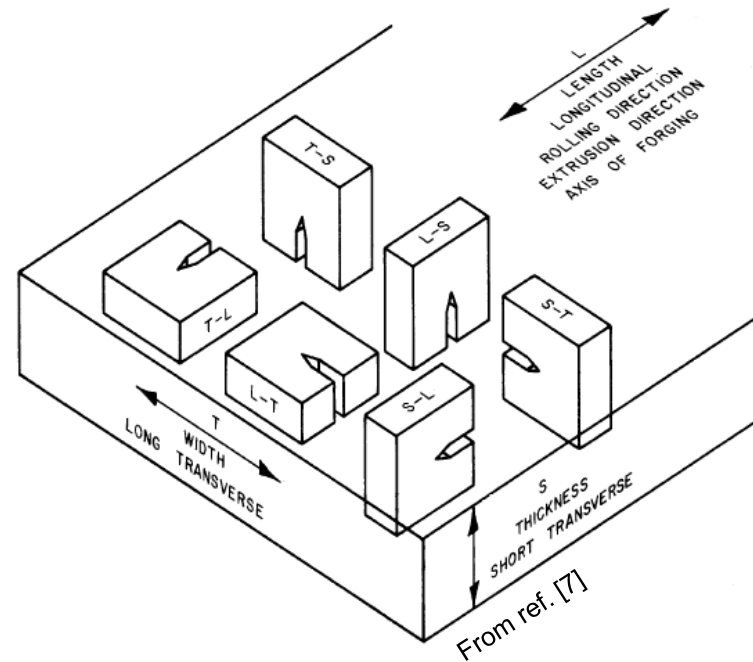
$$K_c = \sigma_{test\ failure} \cdot \beta(a) \cdot \sqrt{\pi \cdot a}$$



Factors / Parameters Affecting Fracture Toughness

Fracture toughness K_{Ic} depends on:

- The material, heat treatment, product form
- Temperature
- Environmental conditions
- Crack orientation and load direction

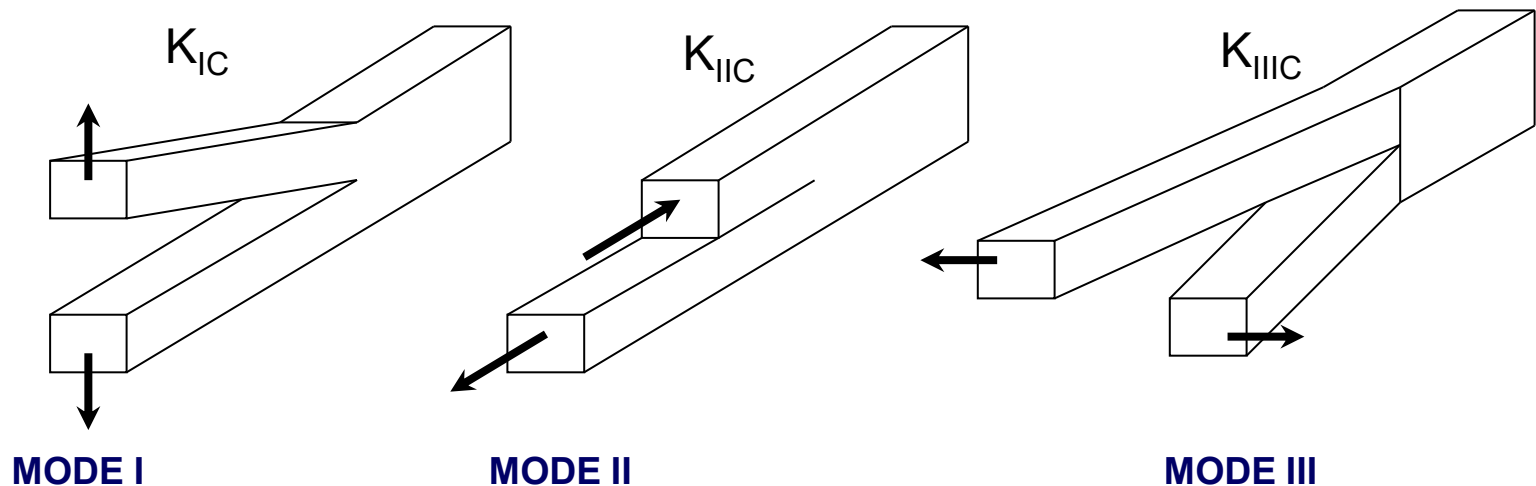


Factors / Parameters Affecting Fracture Toughness

Fracture toughness K_c depends on:

- The material, heat treatment, product form
- Temperature
- Environmental conditions
- Crack orientation and load direction
- Loading mode

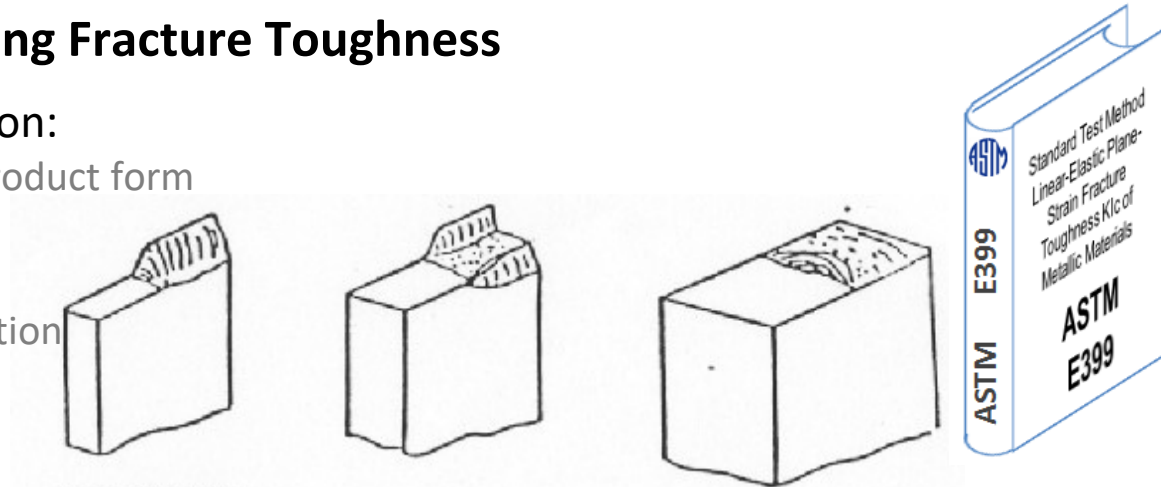
K_c depends not only on material factors / parameters.
Stress-strain field around crack tip depends on
loading mode, so also fracture toughness.
(Similar with other static properties).



Factors / Parameters Affecting Fracture Toughness

Fracture toughness K_{Ic} depends on:

- The material, heat treatment, product form
- Temperature
- Environmental conditions
- Crack orientation and load direction
- Loading mode
- Thickness (stress state)

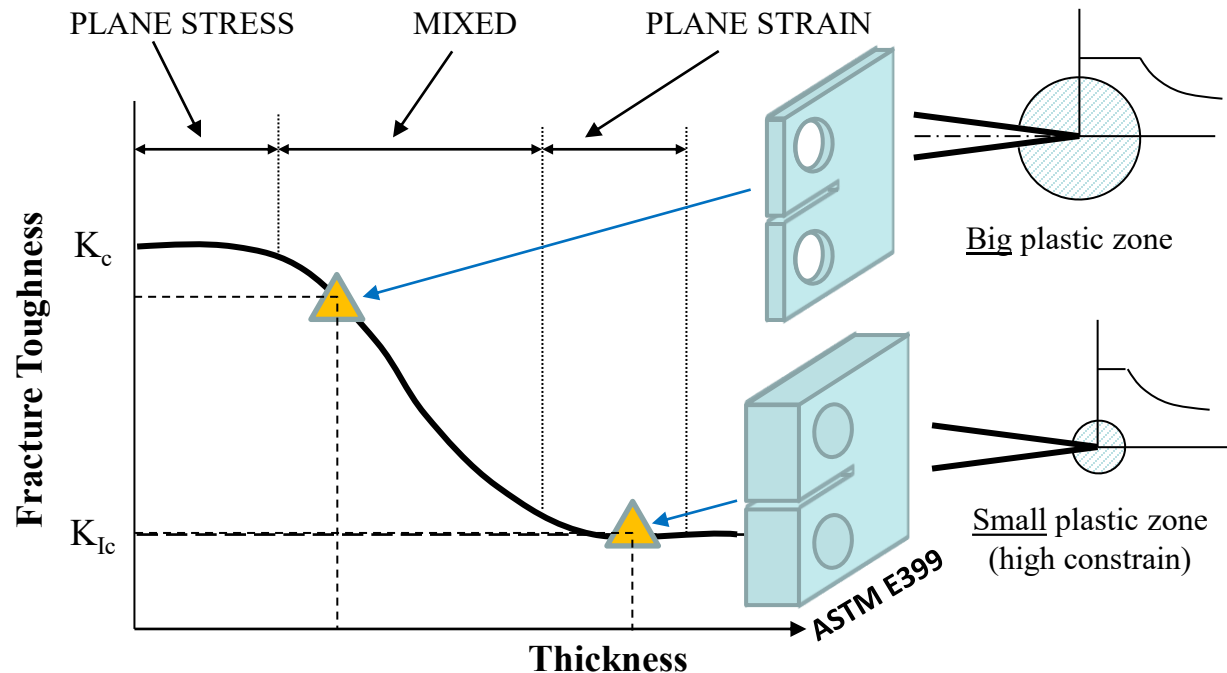


Fracture toughness depends on specimen thickness. Why?

Stress state imposes different constrain conditions on the crack tip.

Plane strain implies high constrain, leading to smaller plastic zone and lower toughness.

Plane stress implies less constrain, leading to greater plastic zone and higher toughness.

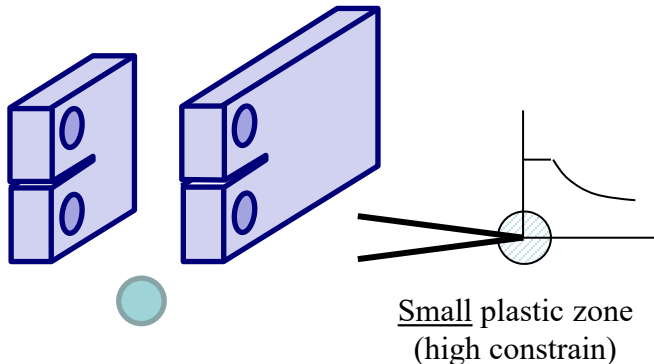


Factors / Parameters Affecting Fracture Toughness

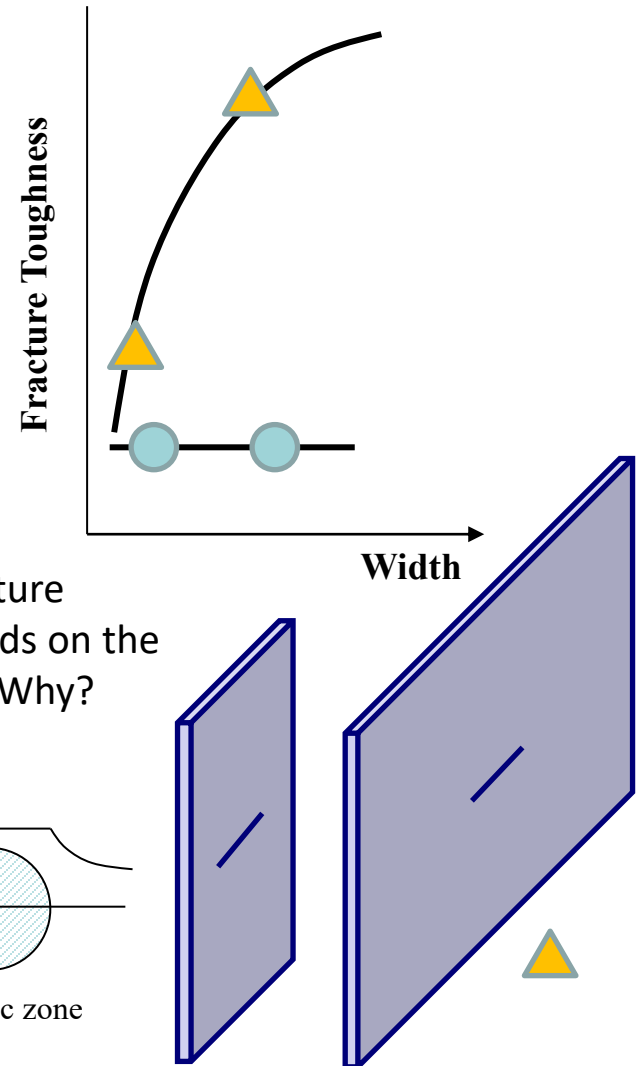
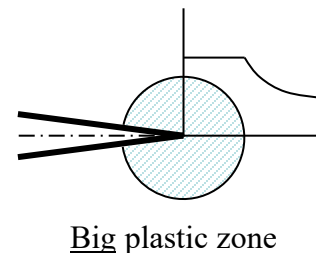
Fracture toughness K_{IC} depends on:

- The material, heat treatment, product form
- Temperature
- Environmental conditions
- Crack orientation and load direction
- Loading mode
- Thickness (stress state)
- Width (plasticity)

Plane strain: fracture toughness does not depend on the specimen width.



Plane stress: fracture toughness depends on the specimen width. Why?



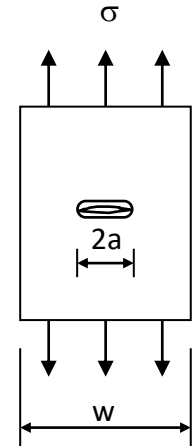
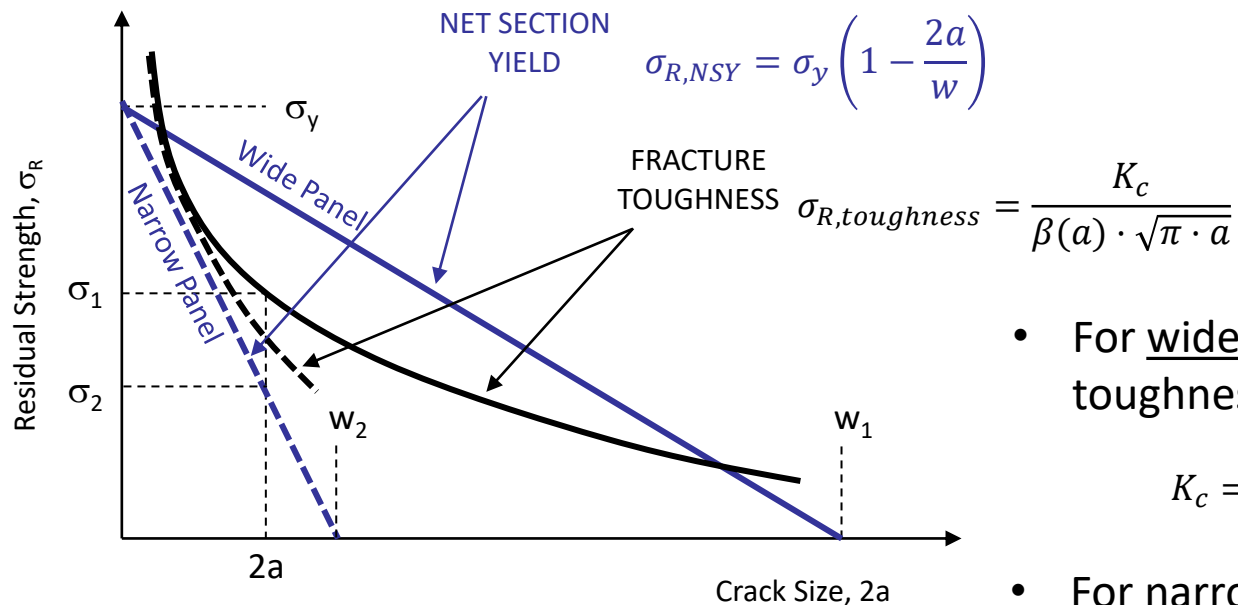
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Net Section Yield vs Fracture Toughness

With a slot there would be static failure, when net section yields.
With a crack it has to be considered additionally fracture toughness.



- For wide panel the fracture toughness criterion is more critical

$$K_c = \sigma_{test\ failure} \cdot \beta(a) \cdot \sqrt{\pi \cdot a}$$

- For narrow panel the net section yield criterion is more critical

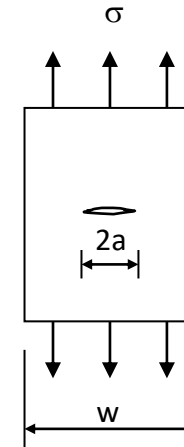
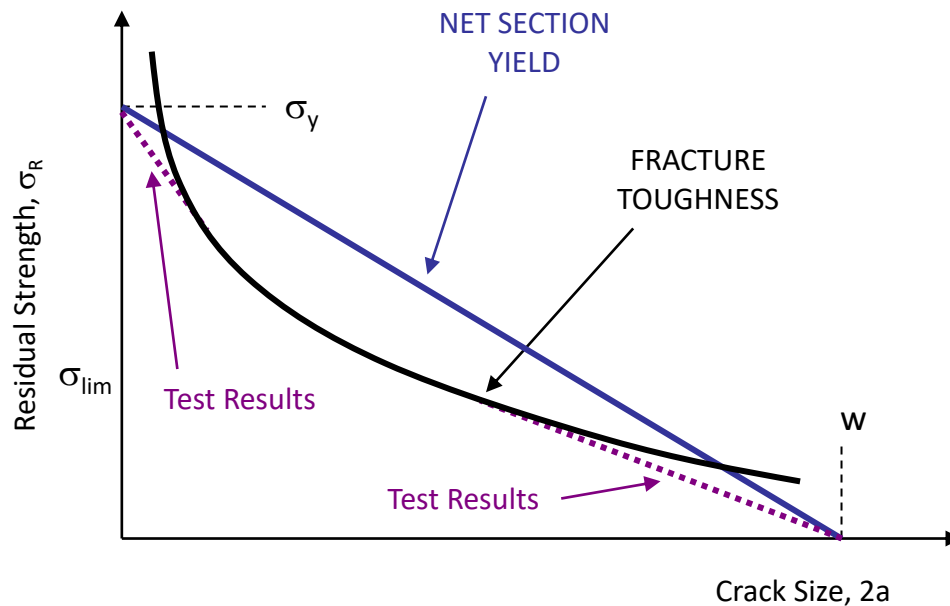
$$K_c^{narrow} = \sigma_{test\ narrow} \cdot \beta(a) \cdot \sqrt{\pi \cdot a}$$

$$K_c^{narrow} < K_c$$

Fracture toughness measurement of thin sheets in plane stress requires wide panels.

Feddersen's Approach

At both sides of the curve, test results falls in the tangency lines, as shown in the sketch.



This is called the Feddersen's approach (Ref. [11]).

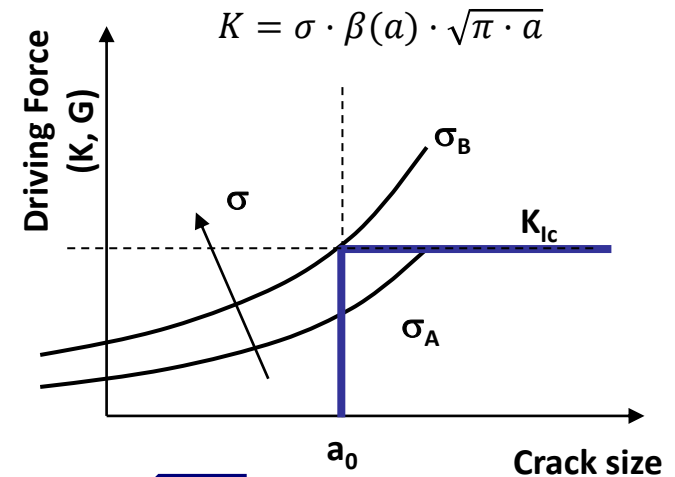
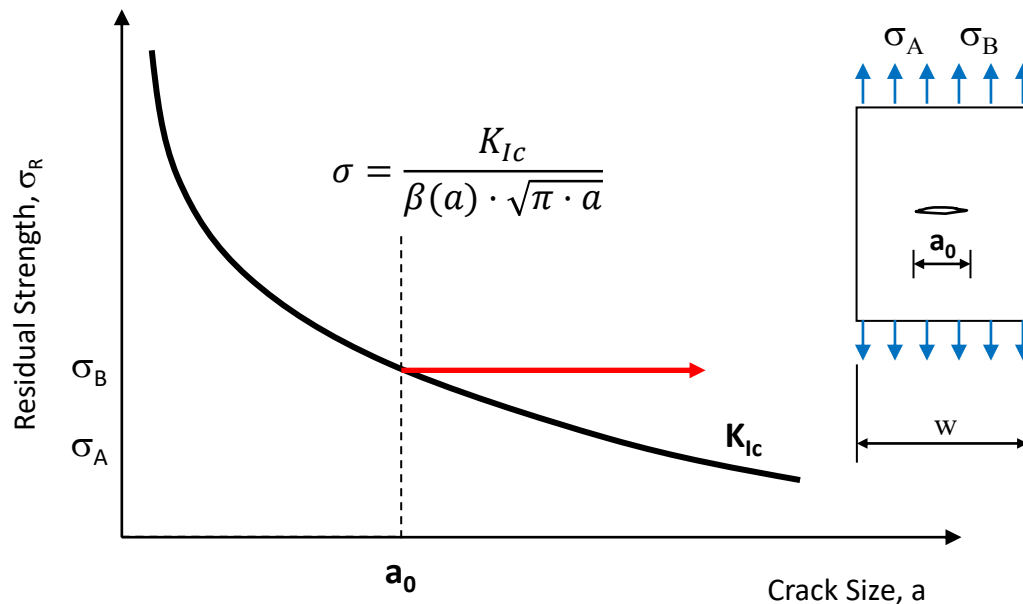
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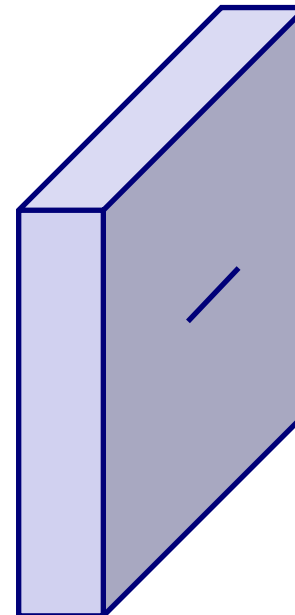
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Plane Strain State

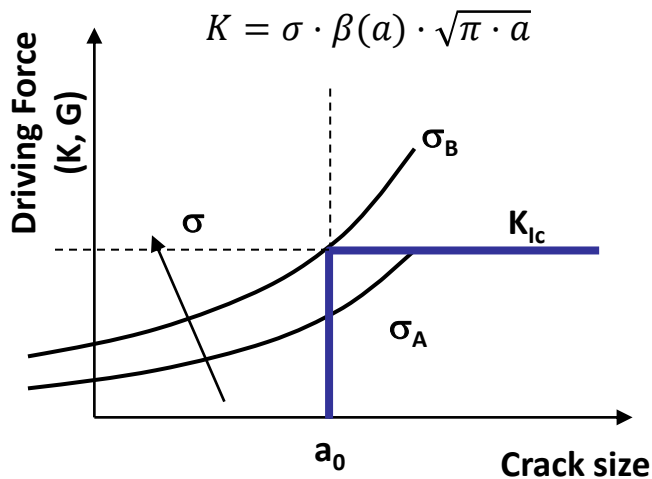
Fracture toughness test under increasing load:



Crack resistance curve depicts the resistance of the material to the applied driving force.



Plane Strain State

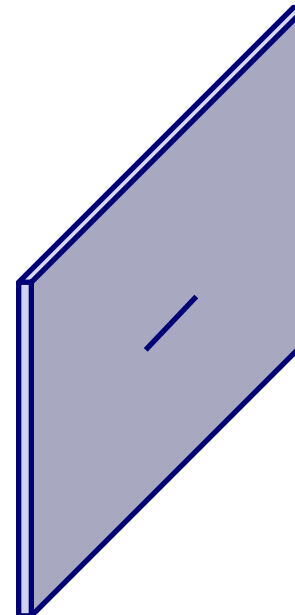
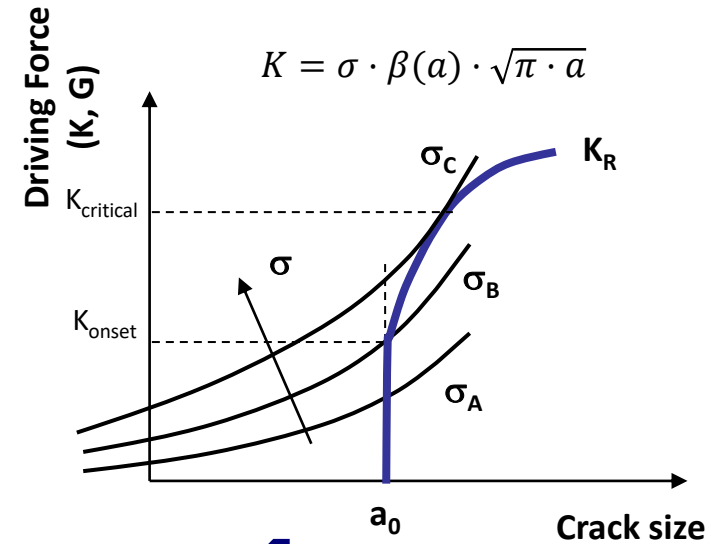
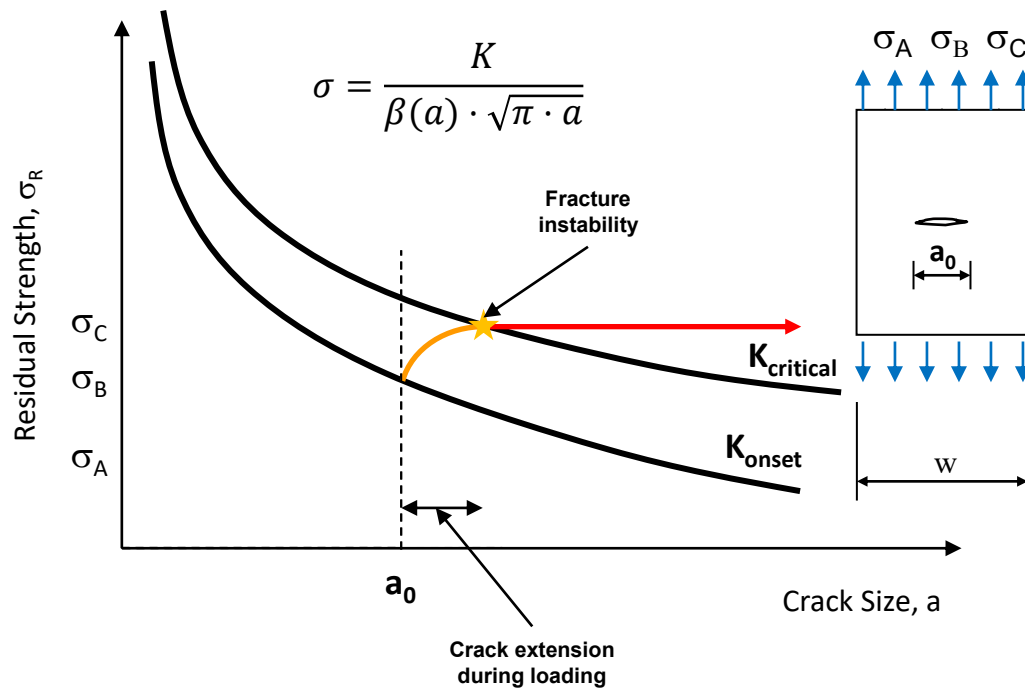


In plane strain test, fracture shows brittle appearance, the resistance curve has a flat shape, failure occurs when $K > K_{Ic}$.

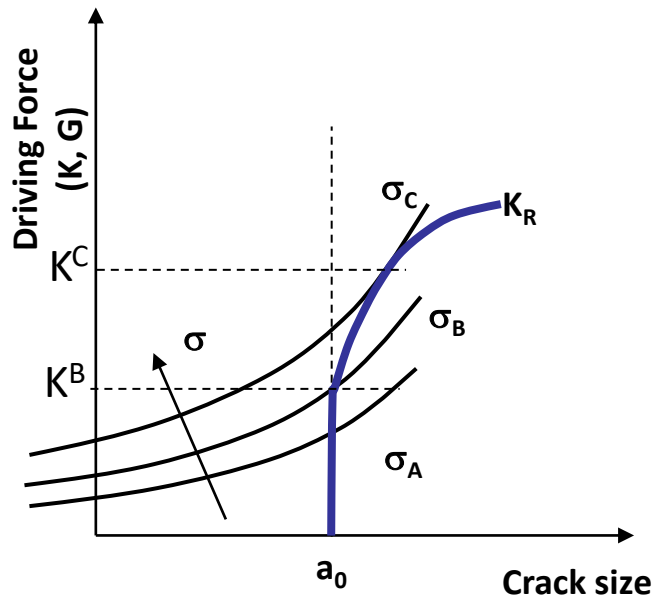
For crack size a_0 , the specimen under driving force associated to K^A will not fail. Residual strength will be determined from K^B .

Plane Stress State

Fracture toughness test under increasing load:

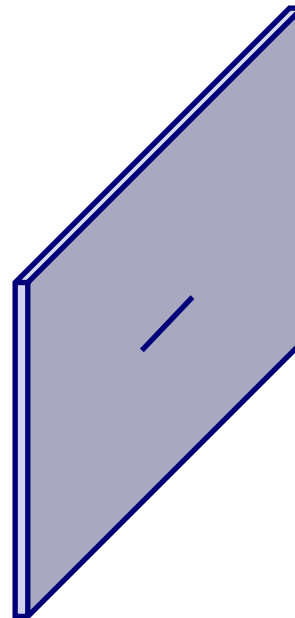


Plane Stress State



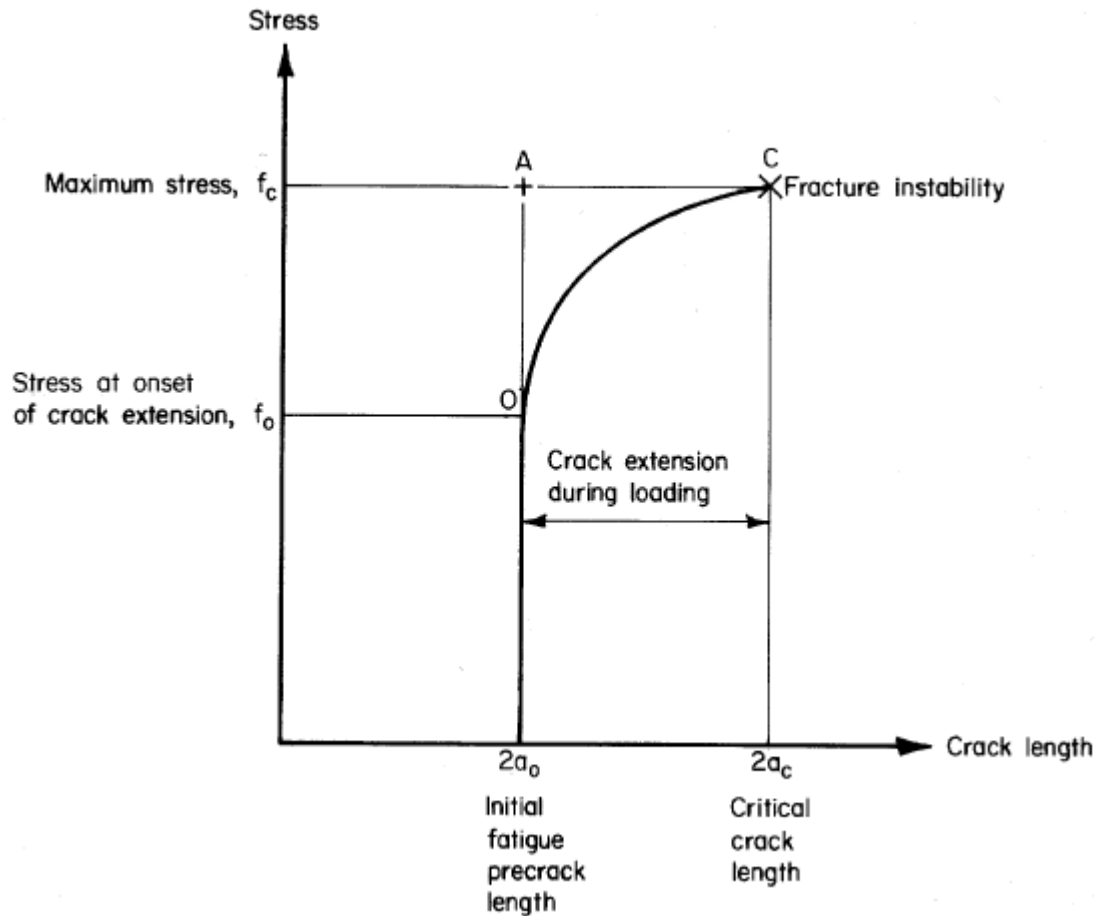
In plane stress test, fracture shows ductile appearance, the resistance curve has a rising shape.

Driving force is spent on crack formation and on plastic deformation, requiring more driving force to continue crack extension up to final rupture. Failure occurs when $K > K_R$.



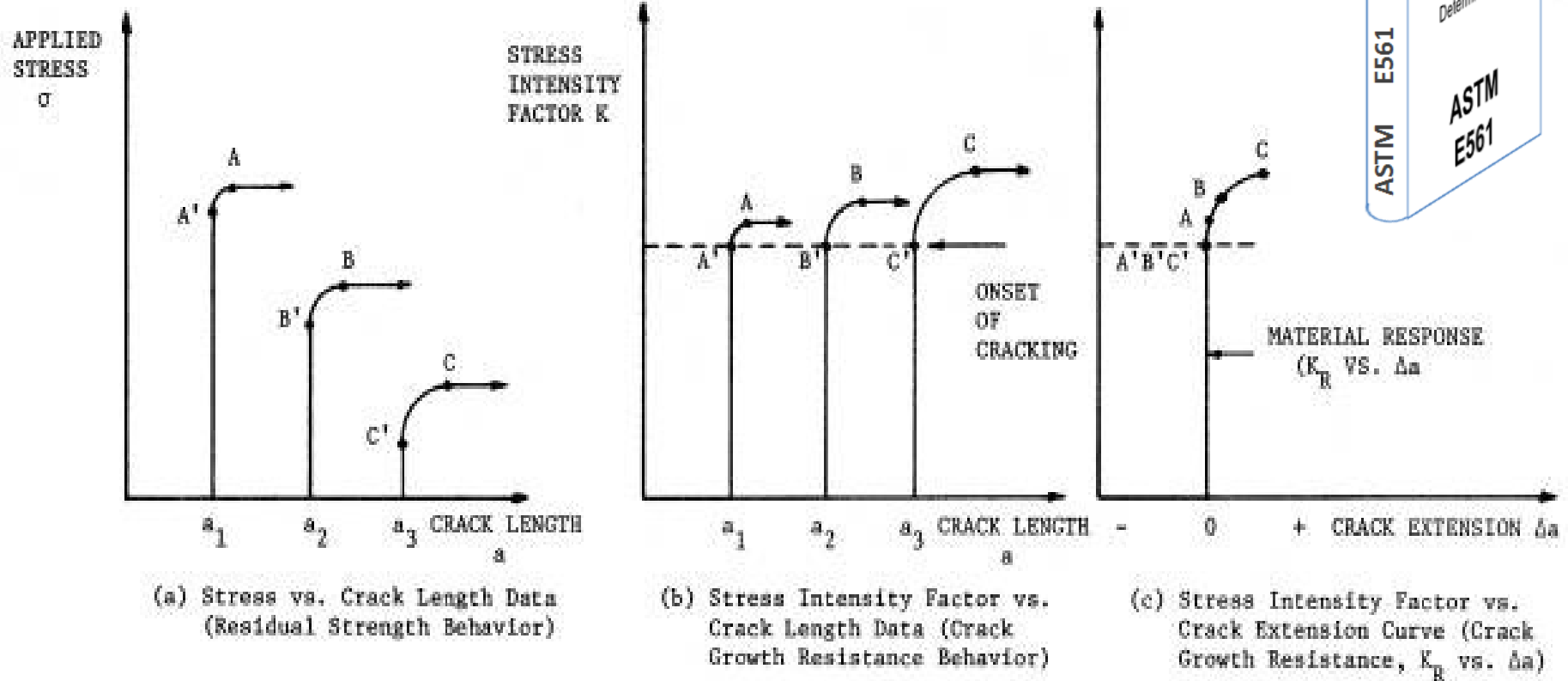
For crack size a_0 , the specimen under driving force σ_B associated to K^B will not fail. Residual strength σ_C will be determined from tangency with K_R curve associated to K^C .

Nomenclature

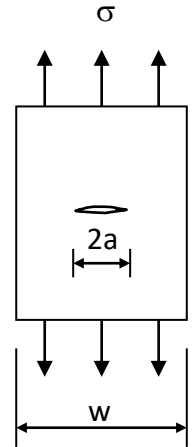
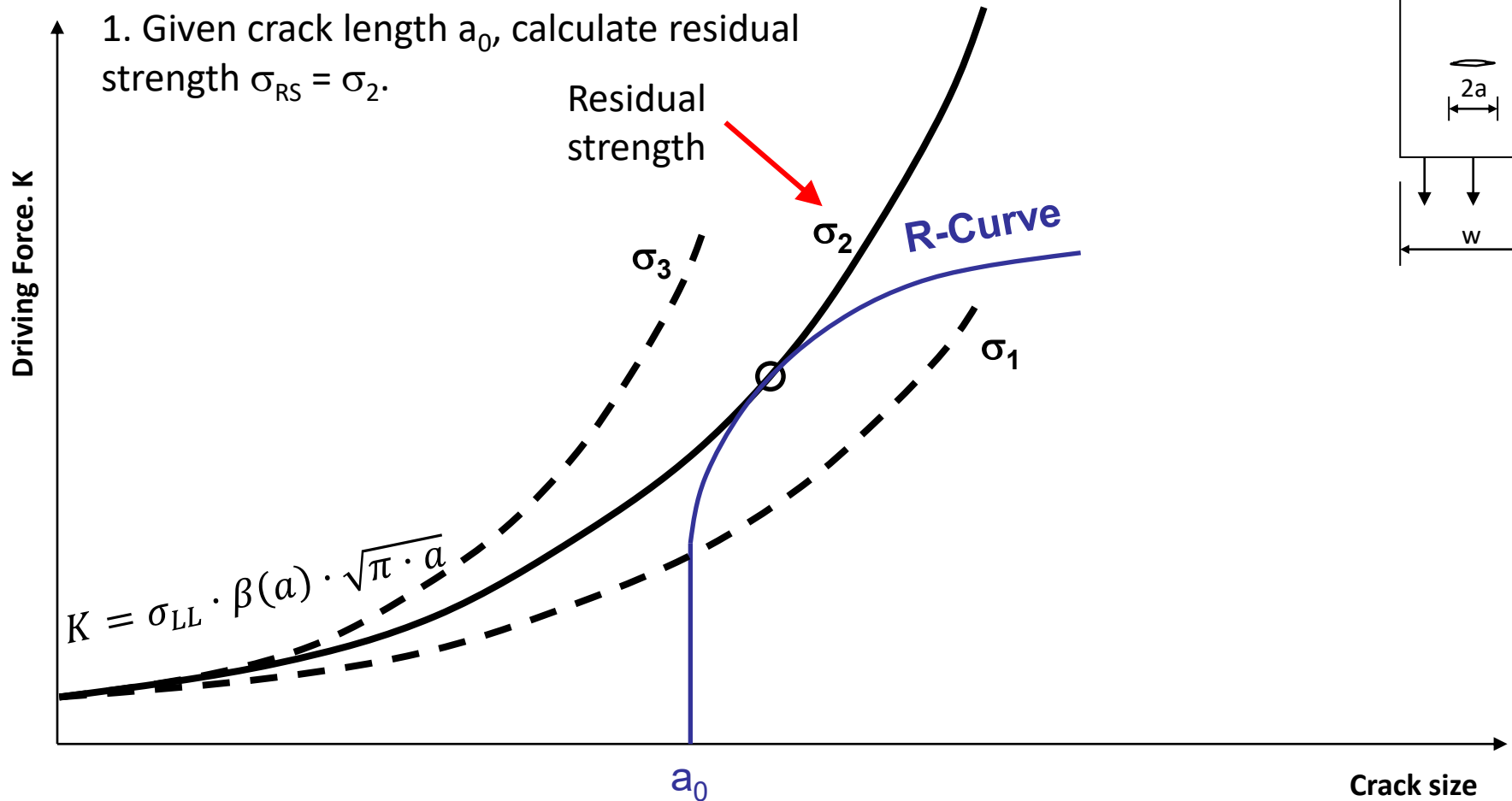


From ref. [7]

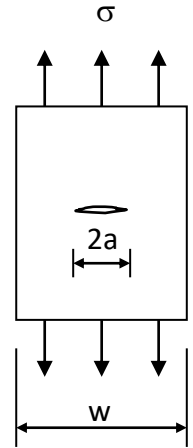
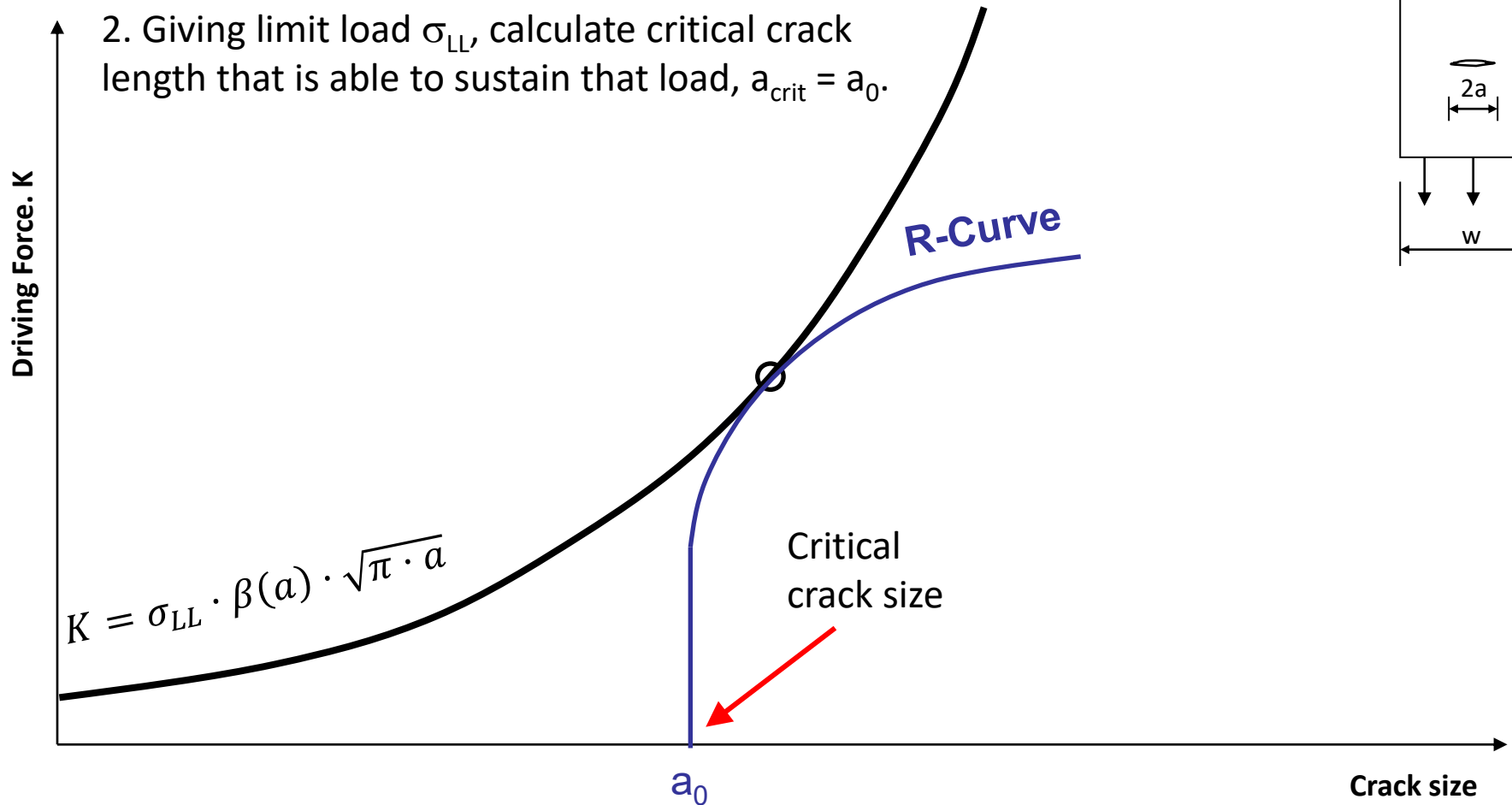
Building the R-Curve



Using the R-Curve



Using the R-Curve



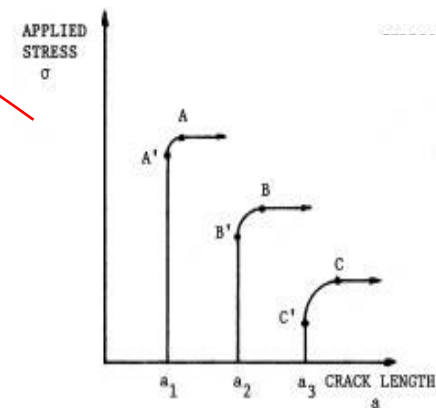
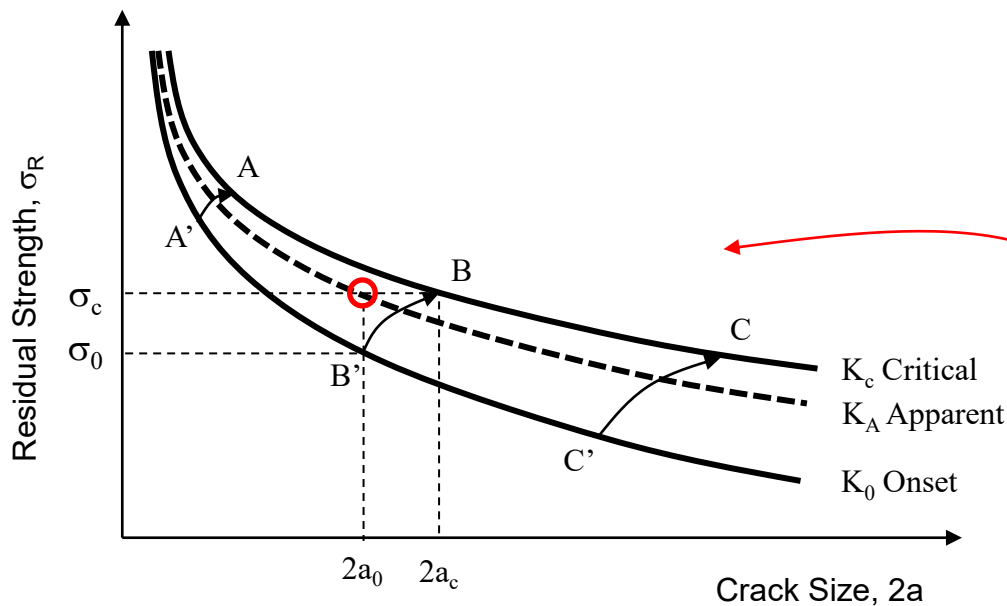
Apparent Fracture Toughness

In the plot, σ_c is the residual strength for crack length $2a_0$, not for $2a_c$.

NASGRO expression for K_c is typically considered representative of the apparent fracture toughness:

$$K_c / K_{Ic} = 1 + B_k e^{-(A_k t/t_0)^2}$$

$$t_0 = 2.5 \left(K_{Ic} / \sigma_{ys} \right)^2$$



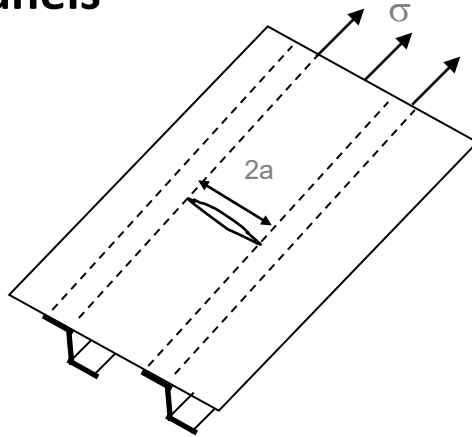
(a) Stress vs. Crack Length Data
(Residual Strength Behavior)

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Stiffened Panels

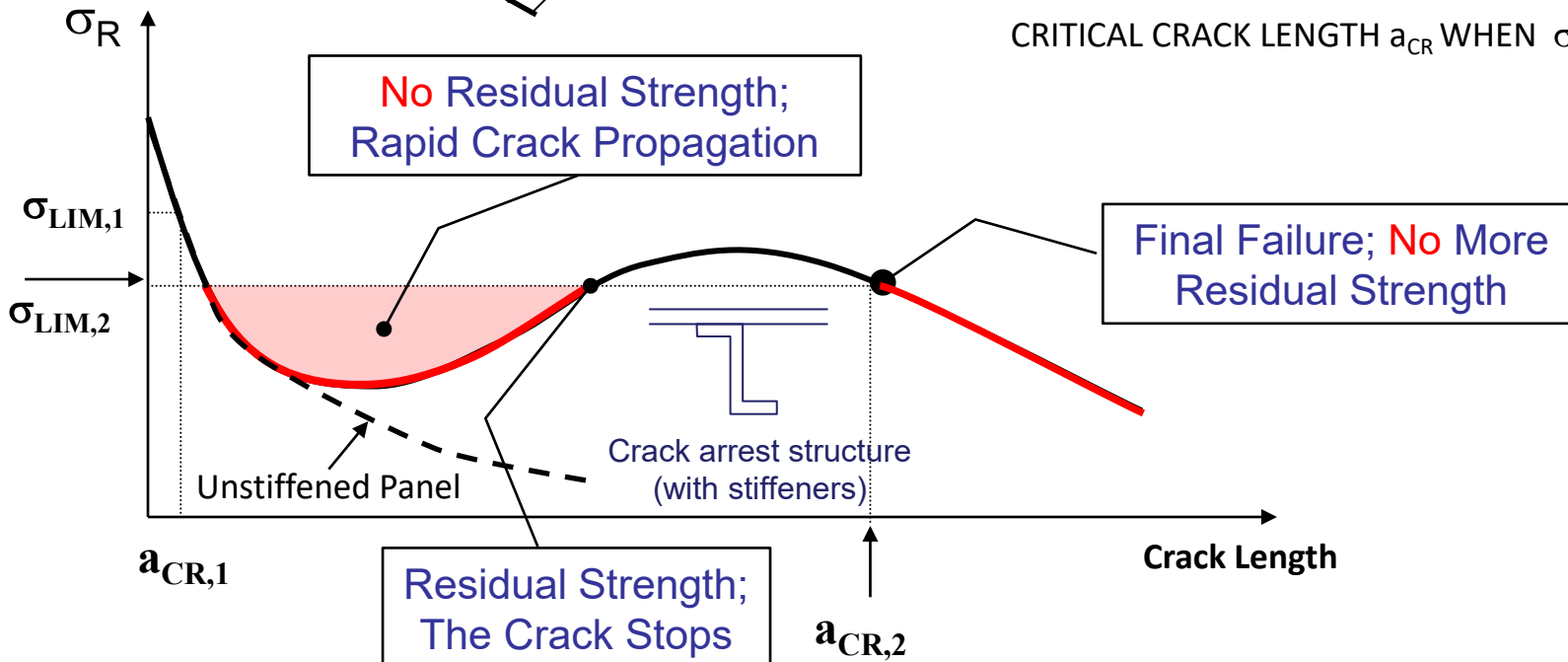


$$\text{Stress Intensity Factor, } K = \beta \cdot \sigma \cdot \sqrt{\pi \cdot a}$$

WHEN $K = K_c \Rightarrow$ RESIDUAL STRESS σ_R CURVE

$$\sigma_R = \frac{K_c}{\beta \cdot \sqrt{\pi \cdot a}}$$

CRITICAL CRACK LENGTH a_{CR} WHEN $\sigma_R = \sigma_{LIM}$

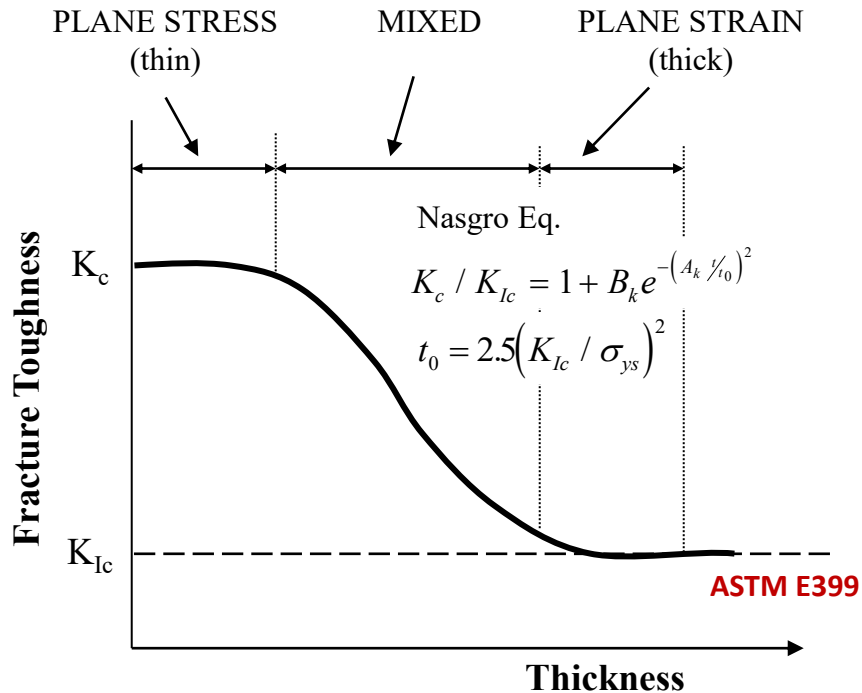


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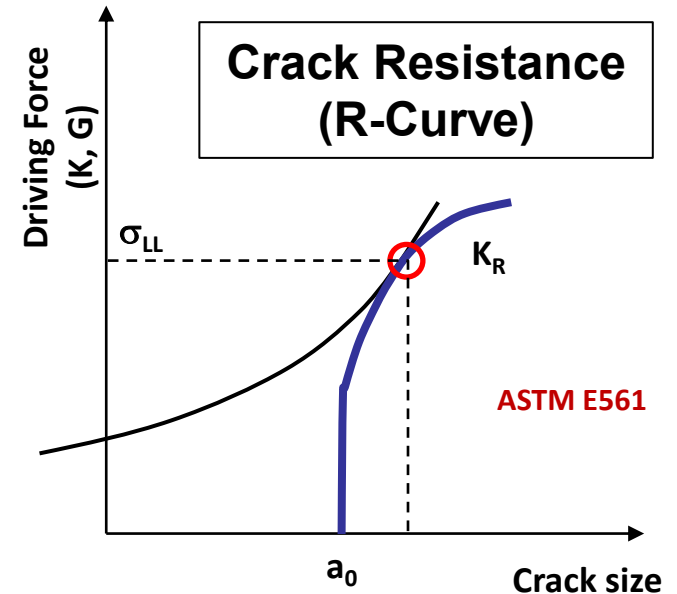
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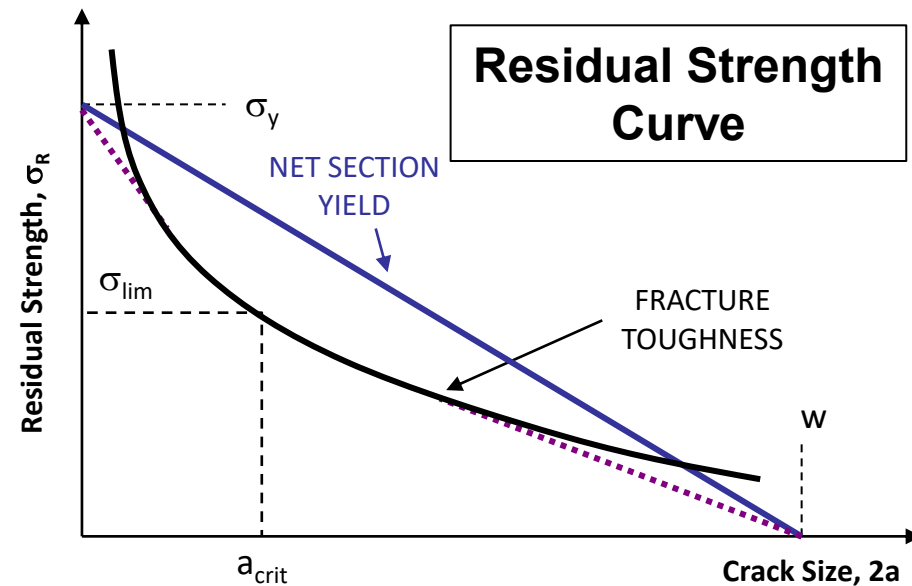
Fracture Toughness



Crack Resistance (R-Curve)



Residual Strength Curve

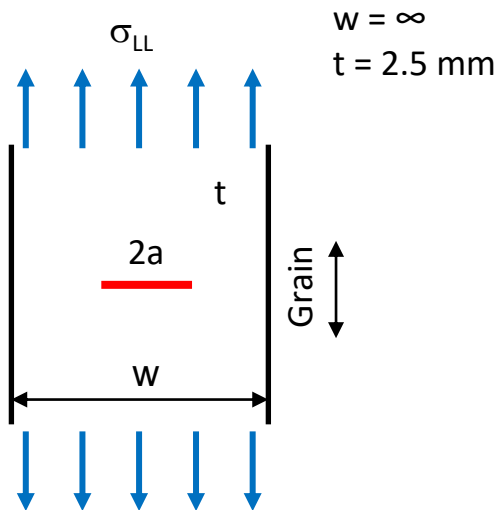


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1. Configuration



PROBLEM

Using the average K_c from Nasgro eq.:

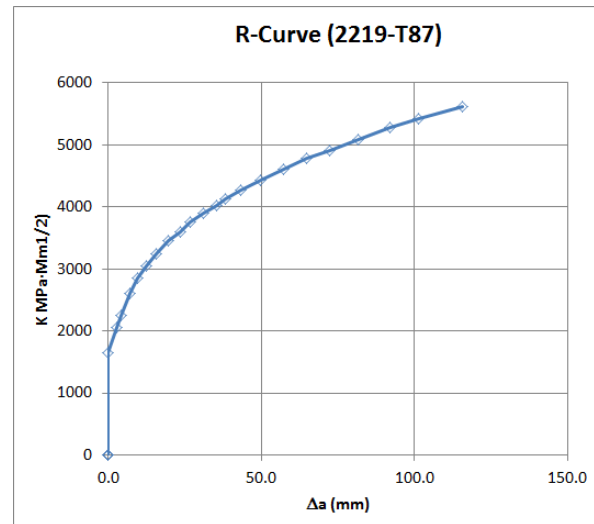
1. What is the stress state
2. Obtain the residual strength curve
Determine the critical crack size:
3. Using average K_c
4. Using R-Curve, if applicable

2. Material

Al 2219-T87 Sheet

From Ref. [8] (MPa , $\text{MPa}\cdot\text{mm}^{1/2}$)

Orientation	F_{ty}	K_{Ic}	A_k	B_k
L-T	393	1042	1	1
T-L	400	938	1	1



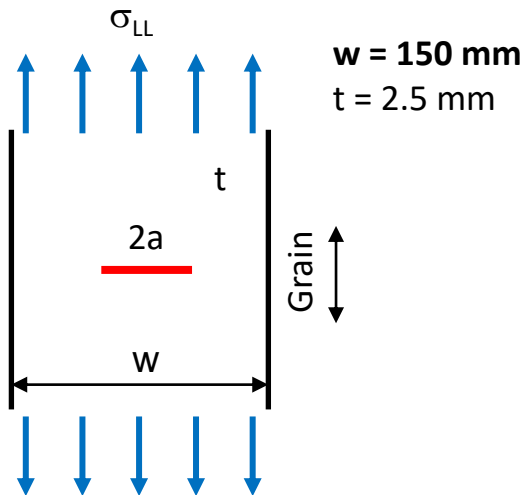
3. Loading

$\sigma_{LL} = 250 \text{ MPa}$

R-Curve (Ref. [12]),
 $t = 2.5 \text{ mm}$

$\Delta a \text{ (mm)}$	$K \text{ (MPa}\cdot\text{mm}^{1/2})$
0.0	0
0.0	1650
2.7	2050
4.1	2250
6.9	2600
9.5	2850
12.3	3050
15.6	3250
19.6	3450
23.5	3600
26.8	3750
30.9	3900
35.2	4024
38.3	4122
43.4	4267
49.5	4425
57.1	4608
64.8	4778
72.1	4912
81.7	5083
92.0	5278
101.2	5412
115.6	5607

1. Configuration



From Ref. 13: $\beta(a) = \frac{1 - 0.5 \frac{2a}{W} + 0.326 \left(\frac{2a}{W}\right)^2}{\sqrt{1 - \frac{2a}{W}}}$

2. Material

Al 2219-T87 Sheet

From Ref. [8] (MPa, MPa·mm^{1/2})

Orientation	Fty	KIc	Ak	Bk
L-T	393	1042	1	1
T-L	400	938	1	1

3. Loading

$\sigma_{LL} = 250 \text{ MPa}$

R-Curve (Ref. [12]),
 $t = 2.5 \text{ mm}$

Δa (mm)	K (MPa·mm ^{1/2})
0.0	0
0.0	1650
2.7	2050
4.1	2250
6.9	2600
9.5	2850
12.3	3050
15.6	3250
19.6	3450
23.5	3600
26.8	3750
30.9	3900
35.2	4024
38.3	4122
43.4	4267
49.5	4425
57.1	4608
64.8	4778
72.1	4912
81.7	5083
92.0	5278
101.2	5412
115.6	5607

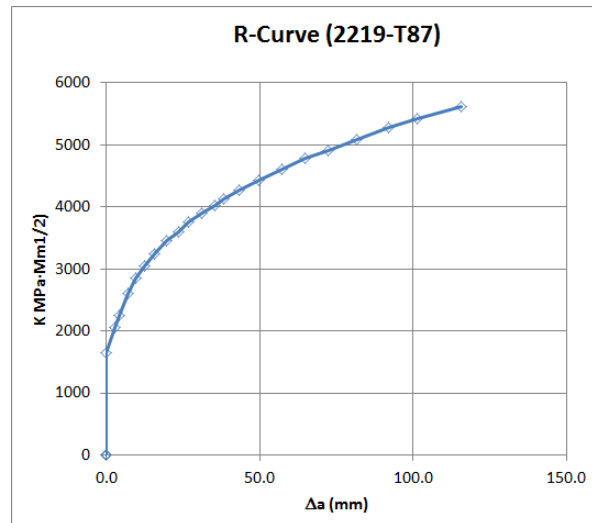
PROBLEM

Similar than exercise 1 but with $W=150 \text{ mm}$

1. Obtain the residual strength and net section yield curve

Determine the critical crack size

1. Using average Kc
2. Using R-Curve, if applicable



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ASTM	American Society for Testing and Materials
β	Geometric Factor
CS	Certification Specification
EASA	European Aviation Safety Agency
FAA	Federal Aviation Administration
FAR	Federal Administration Regulations
G	Energy Release Rate
K	Stress Intensity Factor
K_{app}	Apparent Fracture Toughness
K_c	Fracture Toughness
K_{Ic}	Plane Strain Fracture Toughness
K_R	Crack Resistance Curve
LL	Limit Load
RS	Residual Strength
UL	Ultimate Load

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